

# Mediapipe Human Pose Follow (No-PTZ)

## 1. Experiment Objective

Use MediaPipe Pose to detect 33 human keypoints, with body rotation tracking + distance following keeping the target person centered in the frame and maintaining a constant distance.

## 2. Experiment Principle

### 1. Key Terms

| Term                   | Description                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------|
| MediaPipe Pose         | Google's lightweight human pose estimation that detects 33 human keypoints in real time                |
| Body Center of Gravity | Computes the body reference position from keypoints such as the hip center and shoulder center         |
| Distance Follow        | Estimates distance from the human bounding box width, with PD control maintaining the desired distance |
| Pose Skeleton          | 33 keypoints connected according to the human topology for visualization                               |
| QoS                    | best_effort (image input + cmd_vel downlink), depth=1                                                  |

### 2. Technical Architecture

```
/camera/image_raw (BEST_EFFORT)
  → pose_tracker_node (enable_distance_follow:=true)
    └─ MediaPipe Pose → 33 keypoints → center of gravity + bounding box width
    └─ PD angular velocity + bounding box width distance following → /cmd_vel
```

A MediaPipe Python environment is required (for both ROS2 and the workspace environment).

### 3. Core Topics Quick Reference

#### Firmware Downlink

| Topic    | Message Type        | QoS                        | Field     | Range            | Direction                                |
|----------|---------------------|----------------------------|-----------|------------------|------------------------------------------|
| /cmd_vel | geometry_msgs/Twist | best_effort + keep_last(1) | linear.x  | -0.30~0.30 m/s   | <div>+ = forward, - = backward</div>     |
|          |                     |                            | angular.z | -0.60~0.60 rad/s | <div>+ = left turn, - = right turn</div> |

## 3. Experiment Steps

## 1. Hardware Preparation

| Hardware             | Requirement                       |
|----------------------|-----------------------------------|
| PTZ Version          | ✗                                 |
| No-PTZ Version       | ☑                                 |
| Required Peripherals | ESP32 camera (mounted on bracket) |

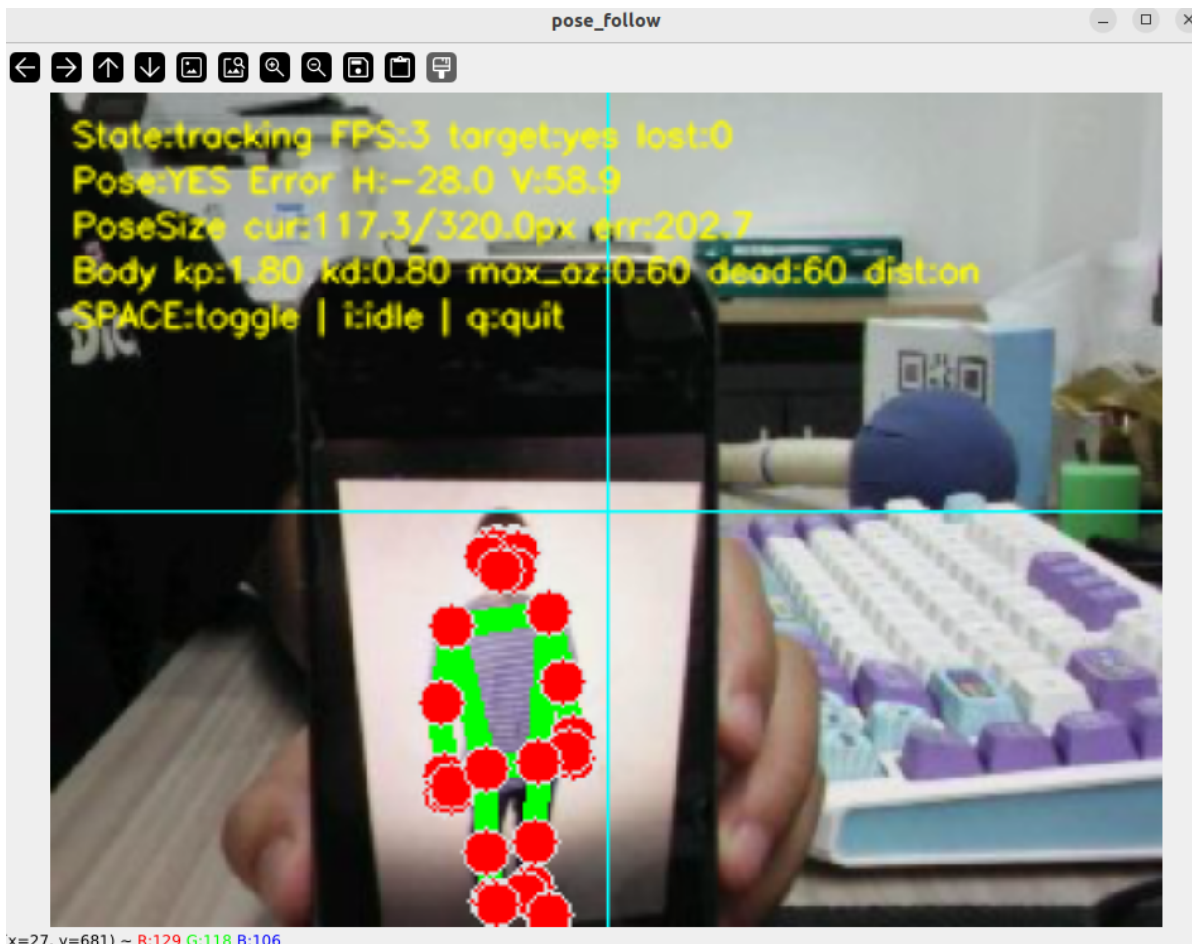
## 2. Start Agent + Camera + Joint Model (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```
yahboom@yahboom: $ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-15-04-46-269642-yahboom-579370
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [579411]
[INFO] [camera_driver-2]: process started with pid [579413]
[INFO] [robot_state_publisher-3]: process started with pid [579415]
[INFO] [joint_state_publisher-4]: process started with pid [579417]
[INFO] [ekf_node-5]: process started with pid [579419]
[micro_ros_agent-1] [1785222287.404236] info      | UDPv4AgentLinux.cpp | init                | running...          | port: 8090
[robot_state_publisher-3] [INFO] [1785222288.847816851] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785222288.848013042] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785222288.848028942] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848037122] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785222288.848043874] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785222288.848050538] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785222288.848057264] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785222288.848063747] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785222288.848070240] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848076888] [robot_state_publisher]: got segment rfwheel_Link
[joint_state_publisher-4] [INFO] [1785222289.372349515] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[camera_driver-2] [INFO] [1785222289.716161193] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[camera_driver-2] [WARN] [1785222292.705741414] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785222294.072701944] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

## 3. Start Human Pose Follow (Terminal 2)

```
ros2 launch microros_v2_no_ptz_visual_control_pkg pose_follow.launch.py
```



## 6. Operation Instructions

| Key     | Function              |
|---------|-----------------------|
| Space   | Toggle follow/standby |
| i / I   | Standby (IDLE)        |
| q / ESC | Exit                  |

## 4. Experiment Result

- Human approaches → the car moves backward; moves away → the car moves forward to follow
- 33 keypoints and skeleton connections displayed in real time
- Human moves out of the frame → stops moving
- With multiple people, the tracking target may switch

## 5. Code Explanation

Source code paths:

- Launch:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/launch/pose_follow.launch.py`
- Node source:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/pose_tracker_node.py`

- Pose detector:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/pose_detector.py`

## 1. Core Node Explanation

`pose_follow.launch.py` launches `pose_tracker_node`, with `enable_distance_follow` defaulting to `true`. Pose detection is computationally heavy (33 keypoints), so image processing defaults to 15 Hz to ensure real-time performance. Distance following is based on the human bounding box width (not radius/size): the wider the bounding box (the closer) → move backward; the narrower (the farther) → move forward.

## 2. Key Parameters

Parameter definition files:

- Launch:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/launch/pose_follow.launch.py`
- Node:  
`~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/pose_tracker_node.py`

| Parameter                                           | Type  | Default                               | Description                         |
|-----------------------------------------------------|-------|---------------------------------------|-------------------------------------|
| <code>pose_detection_confidence</code>              | float | <code>0.5</code>                      | Pose detection confidence threshold |
| <code>pose_tracking_confidence</code>               | float | <code>0.5</code>                      | Pose tracking confidence threshold  |
| <code>deadband_x_pixel</code>                       | float | <code>60.0</code> px                  | Horizontal deadband                 |
| <code>angular_kp</code> / <code>angular_kd</code>   | float | <code>1.8</code> / <code>0.8</code>   | Angle PD gains                      |
| <code>max_angular_z</code>                          | float | <code>0.6</code> rad/s                | Maximum turning speed               |
| <code>enable_distance_follow</code>                 | bool  | <code>true</code>                     | Enable distance following           |
| <code>target_pose_size_px</code>                    | float | <code>320.0</code> px                 | Desired human bounding box width    |
| <code>distance_tolerance_pose_size_px</code>        | float | <code>10.0</code> px                  | Distance tolerance                  |
| <code>distance_kp</code> / <code>distance_kd</code> | float | <code>0.02</code> / <code>0.01</code> | Distance PD gains                   |
| <code>max_linear_x</code>                           | float | <code>0.30</code> m/s                 | Maximum forward/backward speed      |

### 3. Node Description

| Node                           | Package                                            | Description                                                       |
|--------------------------------|----------------------------------------------------|-------------------------------------------------------------------|
| <code>pose_tracker_node</code> | <code>microros_v2_no_ptz_visual_control_pkg</code> | MediaPipe human pose tracking + following, 33 keypoints + dual PD |

### 6. Common Issues

| Issue                             | Cause                                              | Solution                                         |
|-----------------------------------|----------------------------------------------------|--------------------------------------------------|
| Tracking lag                      | High Pose detection computation                    | Lower the image processing resolution or rate    |
| Person not detected               | Too far away or part of the body outside the frame | Keep the whole body within the frame             |
| Interference from multiple people | Multiple people in the frame                       | Make sure only the target person is in the frame |